Homework 11

Solutions will be available after Monday, November 9, 2026.

Question 1: Implementing K-means

K-means is a relatively simple algorithm that we can write ourselves. For Questions 1 and 2, do not use an existing function that performs K-means directly.

Generate 10 observations with two independent standard normal coordinates. Set a random seed and report it. Randomly assign the observations to $K = 2$ clusters, making sure neither cluster is empty. Let $C(i)$ denote the cluster assigned to observation x_i, and let m_k denote the mean of cluster k.

a. Perform one iteration: calculate the cluster means, then assign each observation to its closest mean using Euclidean distance. Report the means and updated assignments, and plot the observations colored by their updated assignments.

b. Write a function that repeats these two steps until the assignments stop changing or a specified maximum number of iterations is reached. Your function should work for a data matrix with any number of variables and a chosen number of clusters K. Start from a nonempty assignment to each cluster; if a cluster becomes empty during iteration, keep its previous mean. Recalculate the cluster means from the final assignments before returning them. Return the final assignments, cluster means, and within-cluster sum of squares,

$$\sum_{k=1}^{K} \sum_{i:\, C(i)=k} \|x_i - m_k\|^2.$$

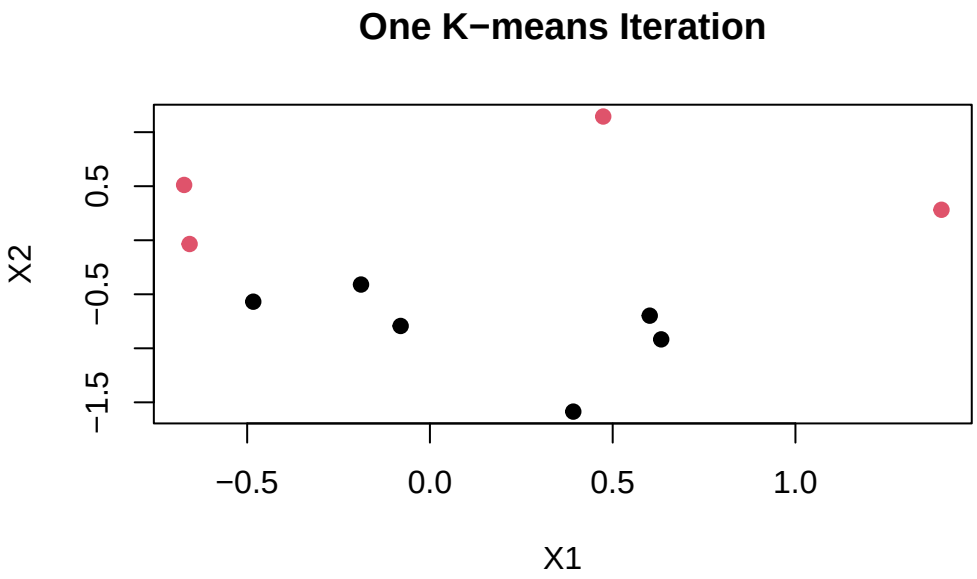
c. Apply your function to the same 10 observations from two different random initial assignments. Report the final plots, cluster means, and within-cluster sums of squares. Do the two runs produce the same grouping and objective value? Explain your findings.

Answer (a)

The observations were generated using seed 432. The initial assignment placed five observations in each cluster, so neither cluster was empty.

Cluster	X1	X2
1	0.1909	-0.8360
2	0.0927	0.2221

Observation	Initial_cluster	Updated_cluster
1	2	2
2	2	2
3	1	1
4	1	1
5	2	2
6	1	1
7	2	2
8	1	1
9	2	1
10	1	1



The initial cluster means were approximately $(0.1909, -0.8360)$ and $(0.0927, 0.2221)$. After one assignment step, observations 1, 2, 5, and 7 were assigned to cluster 2; the remaining observations were assigned to cluster 1.

Answer (b)

The function below checks that the starting assignment is nonempty for every cluster. At each iteration it assigns observations to their nearest current mean and updates means for nonempty clusters; an empty cluster retains its previous mean. It recomputes means from the final assignments before calculating WCSS.

```
my_kmeans <- function(x, K, initial, max_iter = 100) {
  x <- as.matrix(x)
  if (!is.numeric(x) || nrow(x) == 0 || ncol(x) == 0) {
    stop("x must be a nonempty numeric matrix.")
  }
  if (!is.numeric(K) || length(K) != 1 || is.na(K) ||
      K != as.integer(K) || K < 1 || K > nrow(x)) {
    stop("K must be an integer between 1 and the number of rows in x.")
  }
```

```r
  if (!is.numeric(initial) || length(initial) != nrow(x) || anyNA(initial) ||
      any(initial != as.integer(initial)) ||
      any(initial < 1 | initial > K) ||
      !all(seq_len(K) %in% initial)) {
    stop("initial must assign every row to a valid, nonempty cluster.")
  }
  if (!is.numeric(max_iter) || length(max_iter) != 1 || is.na(max_iter) ||
      max_iter < 1 || max_iter != as.integer(max_iter)) {
    stop("max_iter must be a positive integer.")
  }

  groups <- as.integer(initial)
  means <- t(vapply(seq_len(K), function(k) {
    colMeans(x[groups == k, , drop = FALSE])
  }, numeric(ncol(x))))

  for (iter in seq_len(max_iter)) {
    distances <- vapply(seq_len(K), function(k) {
      rowSums(sweep(x, 2, means[k, ])^2)
    }, numeric(nrow(x)))
    new_groups <- max.col(-distances)
    if (identical(new_groups, groups)) break
    groups <- new_groups
    for (k in seq_len(K)) {
      if (any(groups == k)) means[k, ] <- colMeans(x[groups == k, , drop = FALSE])
    }
  }

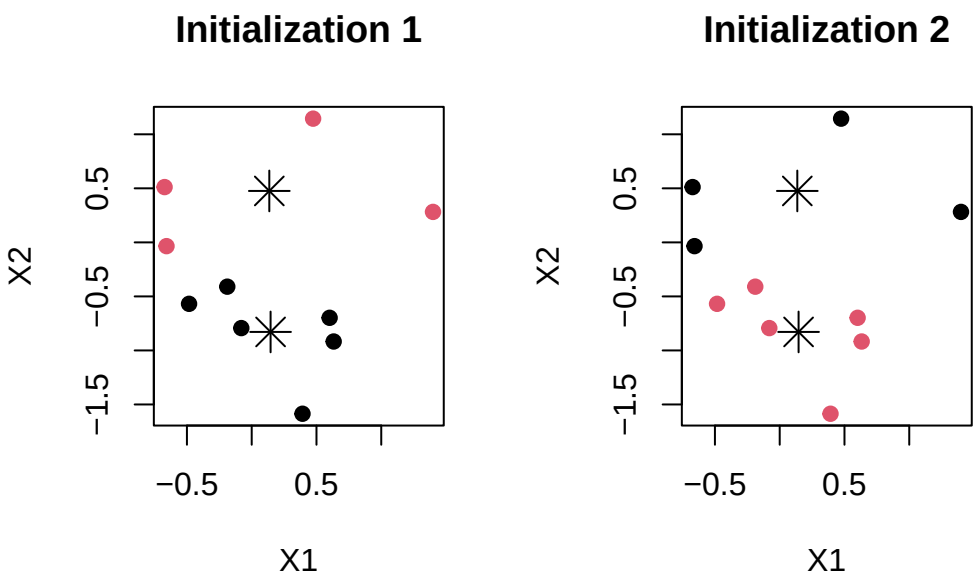
  for (k in seq_len(K)) {
    if (any(groups == k)) means[k, ] <- colMeans(x[groups == k, , drop = FALSE])
  }
  wcss <- sum((x - means[groups, , drop = FALSE])^2)

  list(cluster = groups, centers = means, wcss = wcss)
}
```

Answer (c)

The two runs started from different assignments but converged to the same partition, up to a swap of cluster labels. Both produced a WCSS of **5.6483**. The final groups were observations 1, 2, 5, and 7, and observations 3, 4, 6, 8, 9, and 10. Thus, these two initializations happened to reach the same solution; other initializations can converge to different local optima.

Run	Cluster	X1	X2	WCSS
Initialization 1	1	0.1457	-0.8289	5.6483
Initialization 1	2	0.1359	0.4760	5.6483
Initialization 2	1	0.1359	0.4760	5.6483
Initialization 2	2	0.1457	-0.8289	5.6483



Question 2: Image color compression

Pick your favorite color image and use your K-means function from Question 1 to cluster its pixels. For computational efficiency, resize the image to no more than 200×200 pixels and use its three RGB color channels.

a. Plot the original image and report its dimensions. Transform it into a data matrix with one observation per pixel and three variables: red, green, and blue. Report the dimensions of this matrix.

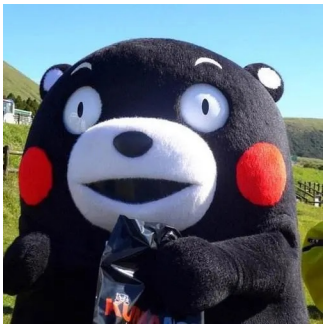
b. For $K = 2, 3, 5, 10$, apply your function to the pixel data. Report the cluster means and within-cluster sum of squares for each K. Reconstruct and plot each compressed image by replacing every pixel with its assigned cluster mean. Use the same RGB scale for all fits.

c. Which value of K would you choose for this image? Briefly justify your choice using the reconstructed images and the within-cluster sums of squares.

Answer (a)

The original Kumamon image is **500 × 500 pixels** with three RGB channels. It was resized to **100 × 100 pixels**, giving a pixel data matrix with **10,000 rows and 3 columns** (red, green, and blue).

Image	Dimensions	Channels
Original image	500 x 500	RGB
Resized image	100 x 100	RGB
Pixel data matrix	10000 x 3	Red, Green, Blue



Answer (b)

I used the `my_kmeans()` function from Question 1, with a different nonempty random initialization for each value of K. All pixel values and cluster means use the same RGB scale from 0 to 1.







The cluster means are the RGB colors used in each reconstruction. The WCSS for each value of K is:

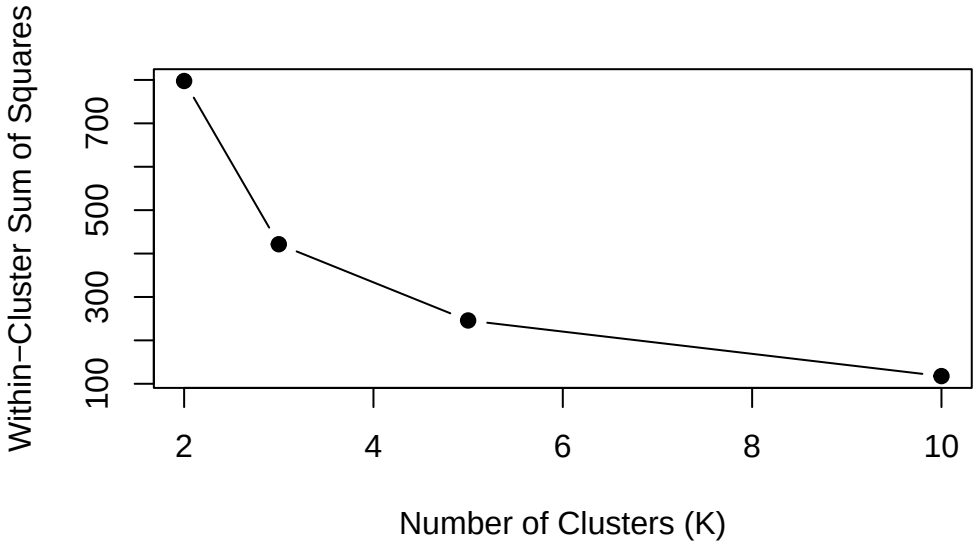
K	Cluster	Red	Green	Blue
2	1	0.170	0.138	0.170
2	2	0.633	0.753	0.818
3	1	0.670	0.867	0.970
3	2	0.514	0.485	0.477

K	Cluster	Red	Green	Blue
3	3	0.122	0.094	0.136
5	1	0.673	0.355	0.203
5	2	0.266	0.262	0.285
5	3	0.529	0.553	0.612
5	4	0.057	0.053	0.105
5	5	0.674	0.880	0.980
10	1	0.589	0.607	0.317
10	2	0.674	0.885	0.985
10	3	0.552	0.592	0.723
10	4	0.258	0.255	0.315
10	5	0.938	0.430	0.374
10	6	0.128	0.121	0.187
10	7	0.679	0.057	0.091
10	8	0.416	0.423	0.479
10	9	0.327	0.322	0.071
10	10	0.034	0.032	0.079

K	WCSS
2	797.58
3	421.50
5	245.85
10	117.68

Answer (c)

I would choose $K = 10$ for this image. The ten-color reconstruction preserves more of the Kumamon's facial shading and the background detail than the two- or three-color versions. It also has the lowest WCSS of the tested choices (117.68), compared with 245.85 for $K = 5$. This improvement comes at the cost of using more colors, so a smaller K could still be reasonable if stronger compression were the priority.



Question 3: Hierarchical clustering of cars

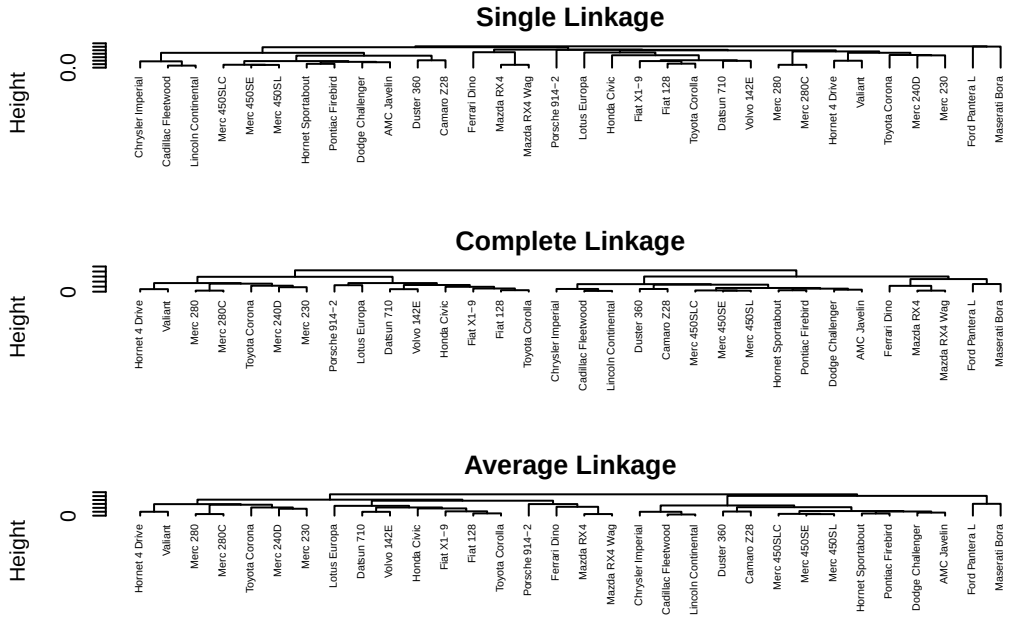
The file `data/mtcars.csv` contains 11 numerical variables for 32 cars, including fuel economy, horsepower, and weight. The `car` column gives the model names; use it to label observations, not as a clustering variable. You may use existing hierarchical clustering functions for this question.

a. Standardize each numerical variable to have mean 0 and standard deviation 1. Using Euclidean distance, perform hierarchical clustering with single, complete, and average linkage. Plot the three dendrograms with the car names and compare the results.

b. Choose one linkage method and a number of clusters that you consider reasonable. Report the cluster memberships and a table of cluster means on the original variable scales. Describe the main characteristics of the groups and explain your choice using the dendrogram and table. Briefly explain why standardizing the variables matters for this analysis.

Answer (a)

I standardized the 11 numerical variables before computing Euclidean distances. Single linkage shows a chaining pattern, adding cars one at a time at relatively short distances. Complete linkage forms more compact groups and has the largest final merge height. Average linkage is intermediate, combining groups more gradually. Thus, the three methods agree on some close pairs but differ in how they join larger groups.

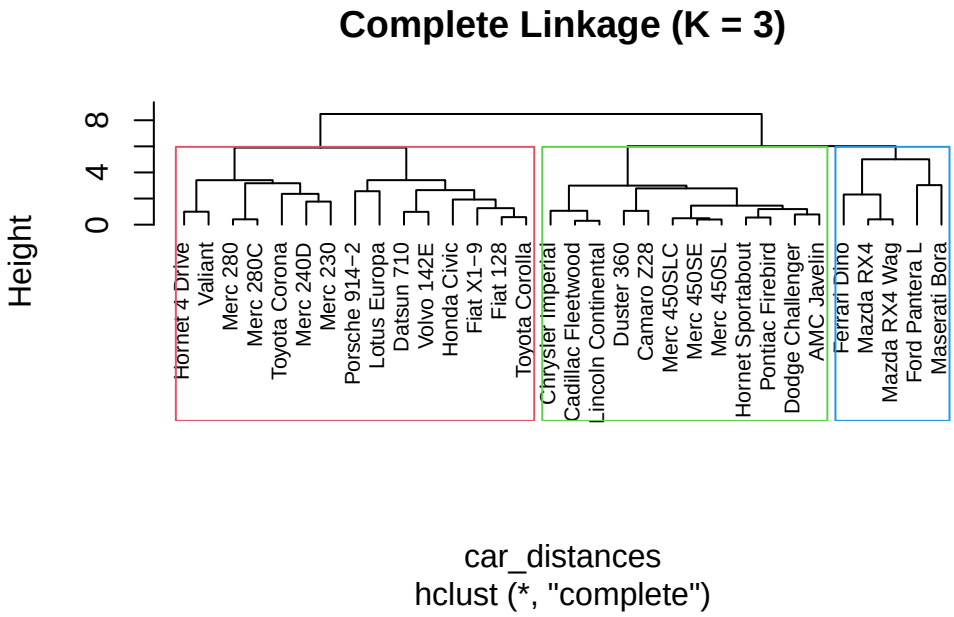


Answer (b)

I chose complete linkage with three clusters. The dendrogram has a large final merge, so cutting it into three groups gives a useful summary before the remaining large groups are joined. The cluster memberships and original-scale means are shown below.

Cluster	Cars
1	Ferrari Dino, Ford Pantera L, Maserati Bora, Mazda RX4, Mazda RX4 Wag
2	Datsun 710, Fiat 128, Fiat X1-9, Honda Civic, Hornet 4 Drive, Lotus Europa, Merc 230, Merc 240D, Merc 280, Merc 280C, Porsche 914-2, Toyota Corolla, Toyota Corona, Valiant, Volvo 142E
3	AMC Javelin, Cadillac Fleetwood, Camaro Z28, Chrysler Imperial, Dodge Challenger, Duster 360, Hornet Sportabout, Lincoln Continental, Merc 450SE, Merc 450SL, Merc 450SLC, Pontiac Firebird

Cluster	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
1	18.50	6.80	223.40	198.80	3.84	3.00	15.62	0.00	1.00	4.60	5.20
2	24.65	4.53	131.65	91.33	3.90	2.58	19.16	0.93	0.53	3.93	1.80
3	15.05	8.00	357.62	194.17	3.12	4.10	17.14	0.00	0.00	3.00	3.08



Cluster 2 (15 cars) is generally the more economical group: it averages 24.65 mpg, 4.53 cylinders, and 2.58 thousand pounds of weight. Cluster 3 (12 cars) contains the heavier, eight-cylinder cars, with the lowest average fuel economy (15.05 mpg), the highest average displacement (357.62), and the greatest average weight (4.10 thousand pounds). Cluster 1 is smaller (5 cars) and more mixed: it includes both the Mazda RX4 models and three high-performance cars; its average horsepower is 198.80 and its average gear count is 4.60.

Standardizing matters because the variables use different units and ranges. Without standardization, large-scale measurements such as displacement or horsepower could dominate Euclidean distances, while variables with smaller numerical ranges would contribute less. Scaling gives each variable comparable influence in the clustering.